POINTWISE WEAKLY ASSOCIATED CRITERION FOR A TENSOR PRODUCT OVER A LOCALLY FINITE TYPE MORPHISM FLAT AT THE POINT

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file spaces-flat.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: a surjective etale descent diagram reducing the pointwise weakly associated criterion for algebraic spaces to the scheme case, with the three point conditions translated definition by definition

Editorial note. The label lemma-bourbaki-finite-type-general-base-at-point names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- The definition of weakly associated points of a quasi-coherent module on an algebraic space, verified after surjective etale descent to schemes
- The relative assassin Ass_{X/Y}(F) defined through the fibres over the base
- Flatness of a module at a point of an algebraic space as an etale local condition
- The scheme case: the Bourbaki criterion for weakly associated points of a tensor product over a locally finite type morphism of schemes flat at the point

Source reference. Original lemma 05IH: sources/stacks-chow/tags.txt, source lines 7131–7131. Prerequisite pointer invoked at proof line(s) 729; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original definition 0CV0: sources/stacks-chow/tags.txt, source lines 16120–16120. Prerequisite pointer invoked at proof line(s) 723; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original definition 0CTX: sources/stacks-chow/tags.txt, source lines 16082–16082. Prerequisite pointer invoked at proof line(s) 717; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

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Source reference. Original definition 05VV: sources/stacks-chow/tags.txt, source lines 7564–7564. Prerequisite pointer invoked at proof line(s) 726; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let S be a scheme. Let $f : X \to Y$ be a morphism of algebraic spaces over S which is locally of finite type. Let $x \in |X|$ with image $y \in |Y|$. Let $\mathcal{F}$ be a finite type quasi-coherent sheaf on X. Let $\mathcal{G}$ be a quasi-coherent sheaf on Y. If $\mathcal{F}$ is flat at x over Y, then*

$$x \in \mathit{WeakAss}_X(\mathcal{F} \otimes_{\mathcal{O}_X} f^*\mathcal{G}) \Leftrightarrow y \in \mathit{WeakAss}_Y(\mathcal{G}) \text{ and } x \in \mathit{Ass}_{X/Y}(\mathcal{F}).$$

Proof. Choose a commutative diagram

$$\begin{array}{ccc} U & \xrightarrow{g} & V \\ \downarrow & & \downarrow \\ X & \xrightarrow{f} & Y \end{array}$$

where U and V are schemes and the vertical arrows are surjective étale. Choose $u \in U$ mapping to x. Let $\mathcal{E} = \mathcal{F}|_U$ and $\mathcal{H} = \mathcal{G}|_V$. Let $v \in V$ be the image of u. Then $x \in \mathrm{WeakAss}_X(\mathcal{F} \otimes_{\mathcal{O}_X} f^*\mathcal{G})$ if and only if $u \in \mathrm{WeakAss}_X(\mathcal{E} \otimes_{\mathcal{O}_X} g^*\mathcal{H})$ by Divisors on Spaces, Definition 0CTX. Similarly, $y \in \mathrm{WeakAss}_Y(\mathcal{G})$ if and only if $v \in \mathrm{WeakAss}_V(\mathcal{H})$. Finally, we have $x \in \mathrm{Ass}_{X/Y}(\mathcal{F})$ if and only if $u \in \mathrm{Ass}_{U_v}(\mathcal{E}|_{U_v})$ by Divisors on Spaces, Definition 0CV0. Observe that flatness of $\mathcal{F}$ at x is equivalent to flatness of $\mathcal{E}$ at u, see Morphisms of Spaces, Definition 05VV. The equivalence for $g : U \to V$, $\mathcal{E}$, $\mathcal{H}$, u, and v is More on Flatness, Lemma 05IH. □